

Reversal Sorting

Time limit: 1.00 s

Memory limit: 512 MB

You have an array that contains a permutation of integers $1, 2, \dots, n$. Your task is to sort the array in increasing order by reversing subarrays. You can construct any solution that has at most n reversals.

Input

The first input line has an integer n : the size of the array. The array elements are numbered $1, 2, \dots, n$.

The next line has n integers $x_1, x_2, \dots, x_n$: the contents of the array.

Output

First print an integer k : the number of reversals.

After that, print k lines that describe the reversals. Each line has two integers a and b : you reverse a subarray from position a to position b .

Constraints

- $1 \leq n \leq 2 \cdot 10^5$

Example

Input	Output
4 2 3 1 4	2 1 3 2 3